

1 · Topology
orchestrator-worker — su-
pervisor (centralized) ↔
swarm (decentralized)



2 · Coordinator
decompose → delegate → aggregate



3 · Verifier
an LLM judge reviews out-
puts (generation ≠ review)



4 · Cost gate
is the work *genuinely parallelizable*?

fans out →
multi-agent

shares context →
**single-
threaded agent**

gate on cost: multi-agent $\approx 15\times$
a chat's tokens — *one first-party
datapoint*, not a law